

HyperLEGNN: Hypergraph Label-Element Neural Networks for Low-Resource Chinese Legal Event Detection

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Abstract. Chinese legal event detection requires trigger localization and fine-grained event-type classification from case facts. The low-resource difficulty is mainly label-side. Many event types have few trigger-level annotations, while legally adjacent types share surface expressions and differ in specific behavior elements. Existing detectors improve contextual encoding or use pairwise label relations, but still estimate event classifiers as independent softmax rows or generic label embeddings. We propose HYPERLEGNN, a heterogeneous label-side hypergraph neural network. The hypergraph contains event-label, trigger-pattern, argument-role, behavior-element, support-instance, and None nodes. Schema, support, and confusion hyperedges encode event composition, K-shot supervision, and label adjacency. A relation-aware hypergraph encoder generates event-type classifier weights for trigger-context vectors, and a gated confusion margin separates labels that share triggers or roles. Experiments on LEVEN and five auxiliary event benchmarks compare encoder, event extraction, few-shot, graph, and hypergraph baselines under strict K-shot, type-supervision K-shot, oracle-trigger, and independent confusable-label settings. HYPERLEGNN improves low-resource Macro-F1, Tail Macro-F1, and Confusable Event F1 with moderate but consistent margins.

Keywords: Chinese legal NLP · Event detection · Low-resource learning · Hypergraph neural networks · Label representation learning

1 Introduction

Chinese legal event detection identifies event triggers in case fact descriptions and assigns them to fine-grained legal event types. The task supports structured fact understanding and can provide event-level evidence for legal judgment prediction, case retrieval, statute retrieval, and legal question answering. CAIL2018 studies criminal judgment prediction from fact descriptions [41]. LeCaRD evaluates Chinese legal case retrieval [27]. STARD studies statute retrieval from real legal queries [32]. JEC-QA focuses on legal-domain question answering [46]. Different from these document-level or retrieval tasks, legal event detection must localize the trigger span and distinguish legally adjacent behaviors in the same fact description. LEVEN provides 8,116 legal documents, 150,977 event mentions,

and 108 event types [43], so it exposes both fine-grained labeling and long-tailed supervision.

The main difficulty is event-type supervision scarcity. Legal text is abundant, but trigger-level annotation is expensive because the annotator must decide both span boundary and event type. Frequent labels provide enough instances to train a classifier row, while rare labels may have only a few trigger examples. The difficulty is amplified by legal label composition. Fraud, theft, robbery, and extortion may all involve property transfer, but they differ in deception, secret taking, violence, threat, or voluntary disposal. A fixed softmax classifier treats labels as unrelated indices, so rare labels receive weak parameter estimates and confusable labels are often separated only by sparse examples.

Existing methods address part of the problem. BERT improves contextual representation [11]. RoBERTa refines robust pre-training [24]. MacBERT strengthens Chinese pre-training [9]. ChineseBERT adds glyph and pinyin information [33]. Longformer supports long-context encoding [1]. Lawformer adapts pre-training to Chinese legal long documents [40]. Event extraction models such as DMCNN [5], DyGIE++ [36], OneIE [22], CasEE [30], Text2Event [25], and UIE [26] model triggers, arguments, or structured event outputs. Low-resource methods such as Matching Network [35], ProtoNet [31], PA-CRF [8], P4E [19], prompt-based event detection [37], and MetaEvent [45] reduce sample dependence. However, label semantics, support triggers, argument roles, behavior elements, and confusable neighbors are rarely modeled as one typed label-side structure.

We propose HYPERLEGNN, a label-side heterogeneous hypergraph method for low-resource Chinese legal event detection. It constructs event-type representations from event labels, trigger patterns, argument roles, behavior elements, K-shot support triggers, and the None class. Schema hyperedges encode event composition, support hyperedges inject few labeled triggers, and confusion hyperedges connect adjacent labels constructed from training-only signals. A relation-aware hypergraph encoder generates dynamic classifier weights for trigger contexts. The contributions are as follows.

- We formulate low-resource legal event detection as dynamic event-classifier generation and construct a heterogeneous label-side hypergraph from dataset schemas, training triggers, and label descriptions.
- We design relation-aware hypergraph propagation for schema, support, and confusion hyperedges, allowing rare labels to receive information from shared roles, behavior elements, and support instances.
- We introduce a confusion gate and a margin objective for adjacent legal labels, targeting substitutions among event types that share triggers or roles but differ in behavior elements.
- We evaluate the method under strict K-shot, type-supervision K-shot, and oracle-trigger settings, and compare it with encoder, extraction, few-shot, graph, hypergraph, and long-tail baselines.

2 Related Work

2.1 Graph Neural Networks and Hypergraph Applications

Graph neural networks learn representations from relational structures. GCN uses normalized neighborhood aggregation [18]. GAT assigns attention weights to neighbors [34]. GraphSAGE supports inductive sampling [16]. R-GCN models typed relations in heterogeneous graphs [29]. Recent surveys summarize these architectures and their trade-offs [39]. In NLP, TextGCN builds document-word graphs [44]. Heterogeneous graph learning uses multiple textual node types [17]. Label-aware GCN propagates information between labels and textual units [4]. These methods motivate label-side modeling, but pairwise edges cannot directly express that one legal event type is jointly determined by a trigger pattern, several argument roles, and behavior elements.

Hypergraph learning represents high-order relations by connecting more than two nodes with one hyperedge. Classical hypergraph regularization was formulated by Zhou et al. [48]. HGNN performs neural propagation over the hypergraph incidence matrix [14]. HyperGCN approximates hyperedges with graph expansion [42]. HyperGAT introduces attention into hypergraph-based text classification [12]. HGNN+ improves adaptive hypergraph modeling [15]. AllSet unifies set-function based hypergraph learning [7]. HYPERLEGNN differs from these models by using typed label-side hyperedges for legal event schema, few-shot support, and label confusion, and by generating classifier weights from event-label embeddings rather than using node representations only as features.

2.2 Legal Event Detection and Low-Resource Event Learning

Chinese legal NLP has moved from document-level prediction toward structured fact understanding. Legal judgment prediction benefits from event-level constraints [13]. LEVEN defines legal event detection with fine-grained event types and trigger annotations [43]. DuEE provides a general Chinese event extraction benchmark [20]. FewFC studies Chinese financial events under limited supervision [47]. DocFEE targets document-level Chinese financial event extraction [6]. MAVEN provides a large general-domain English event benchmark [38]. ACE2005 remains a standard English event extraction benchmark [23]. LEGAL-BERT is used for English legal text representation and is included only in auxiliary English settings [3].

Existing low-resource event methods usually use metric comparison, prompt transformation, or prototype learning. They help when each label has few examples, but they still depend on either support-instance similarity or label text alone. Long-tail losses such as focal loss [21], class-balanced loss [10], and LDAM [2] rebalance gradients but do not change the label parameterization. HYPERLEGNN is complementary to these strategies. It changes the event classifier from a fixed label-index matrix to a label-side hypergraph-generated classifier.

3 Method

3.1 Problem Definition and Candidate Trigger Protocol

Let a legal fact description be $x = (w_1, \dots, w_n)$. A candidate trigger is a span $t = (i, j)$. The model predicts $y \in \mathcal{Y} \cup \{\text{None}\}$, where $\mathcal{Y}$ is the event-type set and None indicates that the candidate is not an event trigger. The evaluation distinguishes three protocols. In *strict K-shot*, the candidate generator, support nodes, and event classifier use only the sampled K trigger annotations per event type. In *type-supervision K-shot*, the candidate generator uses all trigger boundaries in the training split, while event-type supervision is limited to K triggers per type. In *oracle trigger*, gold trigger spans are given and only type classification is evaluated. This separation avoids mixing trigger-boundary recall with label-side type classification.

The candidate generator is a BIO-CRF tagger. We use MacBERT for Chinese datasets and RoBERTa for English datasets. In strict K-shot, it is trained only on sampled triggers and sampled non-trigger spans. In type-supervision K-shot, it is trained on all training-set trigger boundaries. The candidate threshold is selected from $\{0.25, 0.30, 0.35, 0.40, 0.45\}$, and the maximum span length from $\{4, 6, 8\}$ Chinese words or $\{3, 4, 5\}$ English tokens. Non-event candidates are sampled at a 2.5:1 None-to-gold ratio. All candidate-mode baselines use the same candidate list.

3.2 Trigger Context Encoder

For a candidate trigger $t = (i, j)$, the input is the document sentence with boundary markers around the span. A pre-trained encoder produces contextual states $\mathbf{H}$. The trigger representation is

$$\mathbf{h}_t = \left[\mathbf{H}_{[\text{CLS}]}; \text{mean}_{k=i}^j \mathbf{H}_k; \mathbf{H}_i; \mathbf{H}_j \right] \mathbf{W}_t. \quad (1)$$

The same encoder and candidate list are used across candidate-classification baselines. This makes the comparison focus on event-type representation and classifier generation.

3.3 Legal Event Hypergraph

The label-side hypergraph is denoted by $\mathcal{H} = (\mathcal{V}, \mathcal{E}, \mathcal{R})$. It contains three relation families: schema, support, and confusion. The node set contains event-label nodes, trigger-pattern nodes, role nodes, behavior-element nodes, support-instance nodes, and the None node. Table 1 reports construction statistics.

Trigger-pattern nodes. Chinese text is segmented with Jieba and English text with spaCy. Stop words are removed using fixed Chinese and English lists. For each event type, candidate patterns are collected from training triggers, trigger heads, and two-token windows around triggers. We score a pattern by $0.6\text{TFIDF} + 0.4\text{PMI}$, keep the top eight patterns per event type, and merge

Table 1. Hypergraph construction statistics. Support nodes are reported as 8-shot/full-training counts.

Dataset	Event nodes	Trigger nodes	Role nodes	Element nodes	Support nodes	Schema edges	Conf. edges
LEVEN	108	684	23	41	864/94,280	108	42
DuEE	65	392	18	31	520/16,742	65	24
FewFC	26	183	19	26	208/7,562	26	13
DocFEE	15	112	23	18	120/4,860	15	8
MAVEN	168	910	21	62	1,344/67,104	168	56
ACE2005	33	244	17	22	264/4,202	33	14

patterns whose Word2Vec cosine similarity exceeds 0.90. Chinese Word2Vec is trained on training documents and CAIL2018 fact descriptions; English Word2Vec uses the corresponding training documents with Wikipedia initialization [28].

Role and behavior-element nodes. Role nodes are read from the dataset schema. For datasets without explicit roles, we use the role inventory provided by the event extraction format after mapping synonymous names. Behavior elements are extracted from label descriptions and trigger clauses by dependency parsing. The action head and its direct object or modifier form a candidate element. A deterministic mapping table normalizes surface variants. For example, fabricate and deceive are mapped to deception, while seize and threaten are mapped to coercive taking only when a property role is present. The table is built from training data and schema descriptions; no development or test labels are used.

Schema, support, and confusion hyperedges. A schema hyperedge connects one event label with its trigger-pattern, role, and behavior-element nodes. A support hyperedge connects one event label with sampled support-trigger nodes. If a label has fewer than K triggers, all available triggers are used. Support-node features are recomputed at the beginning of each epoch with the current text encoder and detached during hypergraph propagation to reduce memory. Confusion hyperedges use only training-split signals. For event labels y_i and y_j , we compute

$$q_{ij} = 0.40q_{ij}^{\text{desc}} + 0.25q_{ij}^{\text{trig}} + 0.20q_{ij}^{\text{role}} + 0.15q_{ij}^{\text{oof}}, \quad (2)$$

where q^{desc} is label-description similarity, q^{trig} is trigger-pattern overlap, q^{role} is role overlap, and q^{oof} is the training-only confusion frequency estimated by out-of-fold prediction. For each label, at most three neighbors with $q_{ij} > 0.55$ are retained. Connected labels form one confusion hyperedge. Hyperedges with more than five labels are split by complete-link clustering.

In K-shot experiments, all confusion-hyperedge statistics are recomputed within each sampled training split. The out-of-fold classifier used for q^{oof} is trained only on the corresponding sampled training instances. Five folds are used when the sampled labels provide enough instances; otherwise, the number of folds is capped by the minimum available support count. No development or test instances are used for confusion-hyperedge construction.

3.4 Relation-Aware Hypergraph Propagation

Each node $v \in \mathcal{V}$ is initialized by a textual representation of its name, description, or trigger context. For relation r , let $\mathbf{H}_r \in \{0, 1\}^{|\mathcal{V}| \times |\mathcal{E}_r|}$ denote the incidence matrix. A propagation layer is

$$\mathbf{Z}_r^{(l+1)} = \mathbf{D}_{v,r}^{-1/2} \mathbf{H}_r \mathbf{W}_{e,r} \mathbf{D}_{e,r}^{-1} \mathbf{H}_r^\top \mathbf{D}_{v,r}^{-1/2} \mathbf{Z}^{(l)} \mathbf{W}_r^{(l)}. \quad (3)$$

Here $\mathbf{D}_{v,r}$ and $\mathbf{D}_{e,r}$ are node-degree and hyperedge-degree matrices, and $\mathbf{W}_{e,r}$ stores relation-specific hyperedge weights. Relation outputs are fused by

$$\mathbf{Z}^{(l+1)} = \text{LN} \left(\mathbf{Z}^{(l)} + \text{GELU} \left(\sum_{r \in \mathcal{R}} \alpha_r^{(l)} \mathbf{Z}_r^{(l+1)} \right) \right). \quad (4)$$

All relation outputs share hidden size 256. Dropout is applied to node representations and incidence entries.

For a confusion hyperedge e , the gate is computed from its event-label members:

$$\mathbf{p}_e = [\text{mean}_{y \in e} \mathbf{z}_y; \text{max}_{y \in e} \mathbf{z}_y; \text{std}_{y \in e} \mathbf{z}_y], \quad g_e = \sigma(\text{MLP}(\mathbf{p}_e)). \quad (5)$$

The gate scales the corresponding confusion hyperedge weight. Its calibration term is

$$\mathcal{L}_{\text{gate}} = \frac{1}{|\mathcal{E}_c|} \sum_{e \in \mathcal{E}_c} (g_e - c_e)^2, \quad (6)$$

where c_e is the mean pairwise training-only confusion score inside e . Pooling makes the gate well-defined for variable-size hyperedges and avoids pairwise computation inside large groups.

3.5 Dynamic Event Classifier

After propagation, each event label and the None node obtain an embedding $\mathbf{z}_y$. The classifier weight is generated by

$$\mathbf{w}_y = \mathbf{W}_o \mathbf{z}_y + \mathbf{b}_o, \quad s(y \mid x, t) = \mathbf{h}_t^\top \mathbf{w}_y. \quad (7)$$

None is a node and uses the same mapping, so it has no separate free classifier row. The training objective is

$$\mathcal{L} = \mathcal{L}_{\text{ce}} + \lambda_1 \mathcal{L}_{\text{conf}} + \lambda_2 \mathcal{L}_{\text{gate}} + \lambda_3 \|\Theta\|_2^2, \quad (8)$$

where $\mathcal{L}_{\text{ce}}$ is class-balanced cross-entropy, and

$$\mathcal{L}_{\text{conf}} = \sum_{y^- \in \mathcal{C}(y^+)} \max(0, \gamma - s(y^+ \mid x, t) + s(y^- \mid x, t)). \quad (9)$$

We select $\gamma = 0.35$, $\lambda_1 = 0.4$, $\lambda_2 = 0.1$, and $\lambda_3 = 10^{-5}$ on the development set. At inference, support features and event-type embeddings are cached. One hypergraph layer costs $O(|\mathbf{H}|d + |\mathcal{V}|d^2)$, and the gate costs $O(\sum_{e \in \mathcal{E}_c} |e|d)$.

4 Experiments

4.1 Experimental Settings

RQ1 evaluates whether HYPERLEGNN improves LEVEN under type-supervision K-shot, strict K-shot, and oracle-trigger protocols. RQ2 examines whether the gains are more evident on tail labels and independently defined confusable-label groups. RQ3 isolates the effect of label descriptions, support instances, schema hyperedges, confusion hyperedges, gate calibration, and margin learning. RQ4 evaluates whether the same design remains useful on auxiliary event benchmarks with different domains and schemas.

LEVEN is the main dataset. DuEE, FewFC, DocFEE, MAVEN, and ACE2005 are used as auxiliary benchmarks. For K-shot evaluation, K is set to 8, 16, 32, and 64, and at most K gold trigger instances are sampled per event type. Labels with fewer than K triggers use all available triggers. Unsampled triggers in sampled documents are removed from the event-type loss and are not used as None examples. All K-shot experiments are averaged over five seeds, and all compared methods use the same sampled splits. Head, medium, tail, and few-shot labels are defined by more than 100, between 20 and 100, between 5 and 19, and fewer than 5 training triggers, respectively.

For baseline comparisons, candidate-mode baselines receive a marked trigger span and output one label in $\mathcal{Y} \cup \{\text{None}\}$. For encoder-comparison rows, the contextual encoder is changed according to the named baseline. For extraction, few-shot, graph, hypergraph, and long-tail comparisons, the candidate list and trigger-context encoder are kept fixed within each dataset so that the comparison focuses on event-type representation and classifier construction. Graph and hypergraph baselines use the same event nodes, trigger-pattern nodes, role nodes, behavior-element nodes, support nodes, and construction files as HYPERLEGNN. Pairwise graph baselines expand each hyperedge into a clique.

4.2 Compared Methods and Metrics

We compare HYPERLEGNN with encoder, event extraction, few-shot, graph, hypergraph, and long-tail learning baselines. Encoder baselines include BERT [11], RoBERTa [24], MacBERT [9], ChineseBERT [33], Longformer [1], Lawformer [40], and LEGAL-BERT [3]. Event extraction baselines include DMCNN [5], DyGIE++ [36], OneIE [22], CasEE [30], Text2Event [25], and UIE [26]. Few-shot baselines include Matching Network [35], ProtoNet [31], PA-CRF [8], P4E [19], prompt tuning [37], and MetaEvent [45]. Graph and hypergraph baselines include TextGCN [44], heterogeneous GAT [17], Label-GCN [4], R-GCN-label [29], HGNN-label [14], HyperGCN-label [42], HyperGAT-label [12], HGNN+ [15], and AllSet-label [7]. Long-tail baselines include focal loss [21], class-balanced loss [10], and LDAM [2].

In strict K-shot, the candidate generator, support nodes, and event classifier are trained only from the sampled K-shot trigger annotations. Candidate

thresholds and maximum span lengths are selected on the validation data available under the same K-shot protocol, and full training or development trigger boundaries are not used for strict K-shot threshold selection. In type-supervision K-shot, all training-set trigger boundaries are used only for candidate generation, while event-type supervision remains limited to the sampled K triggers per type.

We report Trigger F1, Micro-F1, Macro-F1, Tail Macro-F1, Confusable Event F1, and Confusion Error Rate. Confusable Event F1 is computed on schema-level confusable groups defined before model training by two legal NLP annotators. These groups are fixed across all experiments and are not used to construct the training confusion hyperedges. Confusion Error Rate measures the percentage of event-trigger predictions in which the predicted event type differs from the gold type but belongs to the same independent confusable group. Significance is tested against the best non-HYPERLEGNN baseline using paired bootstrap resampling over test documents with 10,000 samples, followed by Holm-Bonferroni correction.

4.3 RQ1: Main Results and Protocol Control

Table 2. Main LEVEN results under the type-supervision K-shot protocol. K-shot columns report Macro-F1. Full-Ma and Full-Mi denote full-data Macro-F1 and Micro-F1. Tail and Conf. denote Tail Macro-F1 and independent Confusable Event F1. (†) indicates significance against the best baseline after Holm-Bonferroni correction.

Family	Method	8-shot	16-shot	32-shot	64-shot	Full-Ma	Full-Mi	Tail	Conf.
Enc.	BERT	40.7±1.3	46.2±1.2	51.8±0.8	57.7±0.7	65.9±0.4	78.2±0.3	54.5±1.1	49.1±1.0
	MacBERT	42.1±1.4	47.8±1.0	53.6±0.9	59.4±0.6	67.4±0.5	79.0±0.4	56.0±0.8	51.2±0.9
	ChineseBERT	42.8±1.1	48.5±0.8	54.0±0.7	59.9±0.5	68.0±0.3	79.4±0.2	56.6±1.0	51.8±0.7
	Lawformer	44.4±1.2	50.1±0.9	55.7±0.6	61.7±0.7	69.2±0.4	80.2±0.3	58.5±0.9	53.4±0.8
Ext.	CasEE	45.6±1.0	51.2±0.8	56.9±0.7	62.3±0.5	69.1±0.4	80.5±0.3	59.2±0.9	54.5±0.8
	Text2Event	46.1±1.1	51.8±1.0	57.5±0.6	62.8±0.6	69.6±0.3	81.0±0.2	60.0±0.7	55.1±0.6
	UIE	47.2±0.9	52.7±0.7	58.4±0.8	63.3±0.4	70.3±0.4	81.5±0.3	60.9±0.8	56.0±0.7
Few.	ProtoNet	47.7±1.3	53.0±1.1	58.5±0.9	63.3±0.7	68.6±0.5	79.9±0.4	60.3±1.0	55.3±0.9
	P4E	49.0±1.0	54.5±0.8	60.0±0.7	64.5±0.5	69.7±0.3	80.8±0.3	61.5±0.8	56.7±0.7
	MetaEvent	49.8±1.1	55.4±0.9	60.8±0.6	65.0±0.6	70.5±0.4	81.2±0.2	62.2±0.7	57.5±0.8
Graph	Label-GCN	48.1±1.0	53.7±0.9	59.5±0.8	64.4±0.5	70.5±0.4	81.1±0.3	61.3±0.9	56.5±0.7
	R-GCN-label	49.0±0.8	54.7±0.7	60.5±0.6	65.2±0.4	71.1±0.3	81.6±0.2	62.3±0.8	57.3±0.6
	LDAM	49.1±1.2	54.6±0.8	60.3±0.7	64.8±0.6	70.3±0.4	80.9±0.3	61.9±0.7	56.4±0.8
Hyper.	HyperGAT-label	50.7±0.9	56.0±0.7	62.0±0.5	65.9±0.4	71.6±0.3	82.0±0.2	63.0±0.7	58.6±0.6
	AllSet-label	51.0±1.0	56.4±0.6	62.4±0.6	66.1±0.5	71.9±0.3	82.2±0.3	63.5±0.6	59.1±0.5
	HGNN-label	51.4±0.8	56.9±0.7	62.7±0.4	66.4±0.4	72.2±0.2	82.4±0.2	63.8±0.5	59.7±0.7
	HYPERLEGNN	53.2±0.7[†]	58.7±0.5[†]	64.3±0.5[†]	67.7±0.3[†]	73.3±0.3[†]	83.0±0.2	65.7±0.6[†]	61.5±0.6[†]

Table 2 shows that label-side hypergraph generation gives the largest gains in sparse-supervision and diagnostic settings. Compared with HGNN-label using the same label-side resources, HYPERLEGNN improves 8-shot Macro-F1 by 1.8 points, full-data Macro-F1 by 1.1 points, Tail Macro-F1 by 1.9 points, and independent Confusable Event F1 by 1.8 points. The Micro-F1 gain is smaller because head labels account for most test instances. Figure 1 summarizes these results over representative baselines.

Table 3 separates candidate recall from event-type assignment. Strict K-shot gives lower candidate recall, and the limited Trigger F1 gain shows that the label-side hypergraph mainly affects type classification rather than boundary detection. Under oracle triggers, the type-classification gain remains 1.6 points.

Under type-supervision K-shot, all candidate-classification methods use the same candidate list, which isolates the intended low-resource label-classification setting.

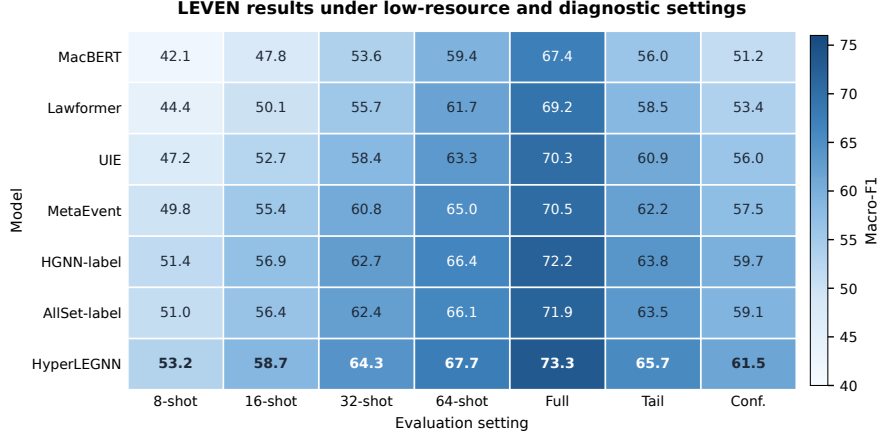


Fig. 1. Main LEVEN result matrix. K-shot and Full columns report Macro-F1. Tail reports Tail Macro-F1. Conf. reports independent Confusable Event F1.

Table 3. Protocol control on LEVEN. Strict K-shot trains the candidate generator and type classifier from K-shot annotations. Type-supervision K-shot uses full training-set trigger boundaries only for candidate generation. Oracle trigger evaluates type classification with gold triggers.

Protocol	Candidate recall	Metric	Lawformer	MetaEvent	HGNN-label	HYPERLEGNN	Gain	95% CI	(p)
Strict 8-shot	67.8±2.1	Trigger F1	50.6±0.8	51.4±0.7	52.1±0.6	52.4±0.5	+0.3	[-0.1,0.7]	.058
Strict 8-shot	67.8±2.1	Macro-F1	37.4±1.4	41.9±1.2	42.8±1.0	44.5±0.9	+1.7	[0.9,2.5]	.011
Type-supervision 8-shot	89.8±0.9	Macro-F1	44.4±1.0	49.8±1.0	51.4±0.8	53.2±0.7	+1.8	[1.0,2.6]	.008
Oracle trigger 8-shot	100.0	Type Macro-F1	50.0±0.9	56.0±0.7	57.2±0.7	58.8±0.6	+1.6	[0.8,2.3]	.014
Full-data candidate	95.7±0.4	Macro-F1	69.2±0.4	70.5±0.4	72.2±0.3	73.3±0.3	+1.1	[0.6,1.5]	.018

4.4 RQ2: Tail Labels and Independent Confusable Groups

Figure 2 reports two diagnostic views. In independent confusable groups, HYPERLEGNN improves Confusable Event F1 by 1.7 to 2.5 points over HGNN-label, with larger gaps on financial-fraud and violence-order groups. The protocol-controlled plot shows that the gain is not tied to one candidate-generation setting. The weakest improvement appears in strict K-shot Trigger F1, while type-level metrics preserve clearer gains.

4.5 RQ3: Ablation Study

Table 4 isolates the main design choices. Replacing generated classifier weights with ordinary softmax rows causes the largest reduction, especially on Tail Macro-F1 and Confusable Event F1. Label-description and support-prototype variants retain part of the performance but do not model schema and support evidence jointly. Removing confusion hyperedges, the gate, or the margin loss

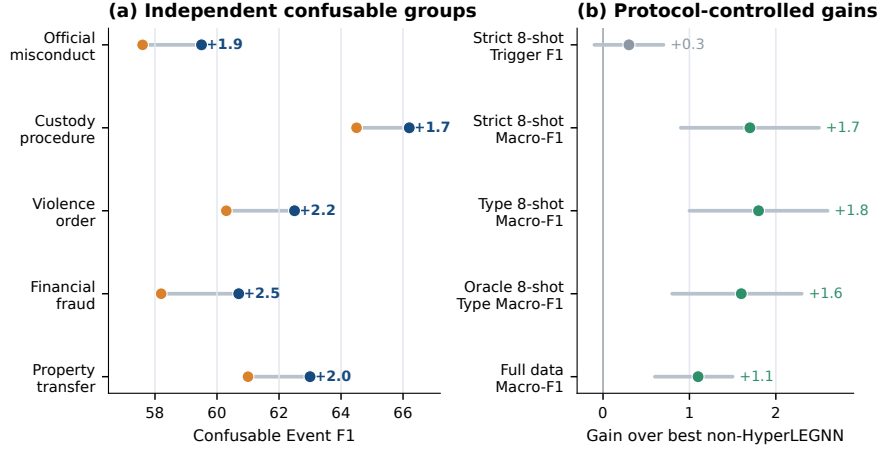


Fig. 2. Additional result diagnostics. Orange points in panel (a) denote HGNN-label and blue points denote HYPERLEGNN. Panel (b) reports gains over the best non-HYPERLEGNN baseline with bootstrap confidence intervals.

Table 4. Ablation study on LEVEN. CER denotes all-label Confusion Error Rate. Lower CER is better.

Variant	8-shot	Full-Ma	Tail	Conf.	CER
Full HYPERLEGNN	53.2±0.7	73.3±0.3	65.7±0.6	61.5±0.6	18.4
Ordinary softmax classifier	49.4±1.3	70.4±0.5	60.8±1.0	56.5±1.1	23.6
Label-description classifier only	50.2±1.1	71.0±0.4	62.3±0.9	58.1±1.0	22.1
Support-prototype classifier only	50.6±1.0	71.2±0.4	62.8±0.8	58.6±0.9	21.8
Schema nodes without propagation	51.0±0.9	71.6±0.4	63.3±0.7	59.0±0.8	21.1
Without schema hyperedges	52.0±0.8	72.5±0.3	64.2±0.7	60.1±0.7	20.0
Without support hyperedges	52.3±0.8	72.8±0.3	64.4±0.6	60.4±0.7	19.7
Without confusion hyperedges	52.2±0.8	72.6±0.3	64.8±0.7	59.7±0.8	20.4
Without confusion gate	52.6±0.7	72.9±0.3	65.1±0.6	60.2±0.7	19.8
Without margin loss	52.7±0.7	73.0±0.3	65.2±0.6	60.5±0.6	19.5
Support refresh every 5 epochs	53.0±0.6	73.2±0.3	65.5±0.5	61.2±0.6	18.6
No-detach support cache	53.3±0.8	73.4±0.4	65.8±0.7	61.6±0.7	18.3

mainly affects Confusable Event F1 and CER, which is consistent with their use for adjacent-label separation. The no-detach support cache produces marginal mean-score changes within the reported variation, but it increases memory cost because support-trigger representations remain attached to the encoder graph. We therefore keep the detached support cache in the adopted model.

4.6 RQ4: Auxiliary Datasets and Error Analysis

Table 5 provides an auxiliary cross-benchmark diagnostic. The gains are smaller than those on LEVEN, especially on document-level or general-domain datasets. On MAVEN and ACE2005, the gain mainly comes from label descriptions, trigger patterns, and support nodes. We further inspect 300 LEVEN errors from the full-data setting, sampled uniformly from head, medium, and tail labels. The

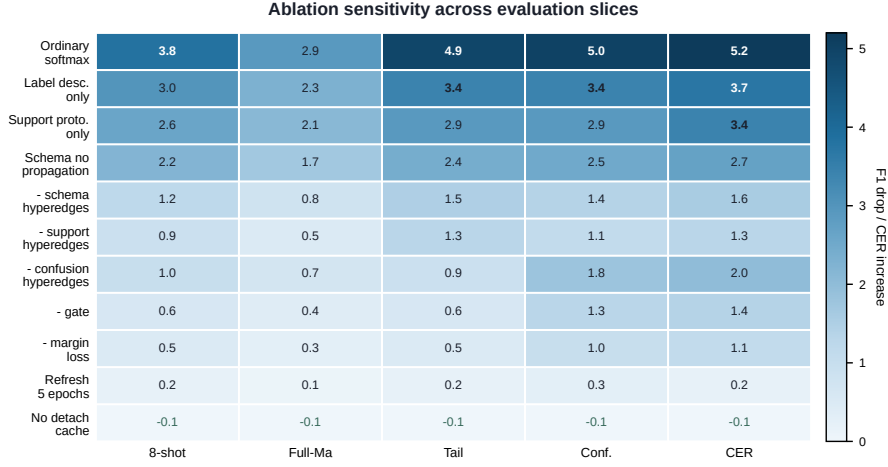


Fig. 3. Ablation sensitivity across evaluation slices. Values denote F1 drop from the full model. The CER column reports the increase in Confusion Error Rate.

Table 5. Auxiliary dataset results. The best baseline is identified separately for each dataset. Scores are full-data Macro-F1.

Dataset	Language / type	Best baseline	Baseline	HYPERLEGNN w/o elements	HYPERLEGNN	Gain
DuEE	Chinese general events	UIE	76.4±0.5	77.1±0.4	77.9±0.3	+1.5
FewFC	Chinese financial events	MetaEvent	71.8±0.4	72.6±0.5	73.4±0.4	+1.6
DocFEE	Chinese document events	Text2Event	67.2±0.6	67.7±0.3	67.9±0.4	+0.7
MAVEN	English general events	OneIE	73.0±0.3	73.7±0.4	74.5±0.3	+1.5
ACE2005	English ACE events	DyGIE++	72.7±0.5	73.3±0.4	73.7±0.5	+1.0

remaining errors are assigned to trigger-boundary ambiguity, missing implicit intent, confusable property-transfer labels, long-distance argument evidence, and annotation inconsistency. Trigger-boundary errors dominate strict K-shot, while type substitution dominates oracle-trigger evaluation. This split is consistent with Table 3 and indicates that label-side modeling mainly addresses type assignment after candidate generation.

5 Conclusion

This paper presents HYPERLEGNN, a heterogeneous label-side hypergraph neural network for low-resource Chinese legal event detection. The method represents each event type through schema components, K-shot support instances, and confusable-label relations, then generates dynamic classifier weights for trigger-context classification. Experiments on LEVEN and auxiliary event benchmarks show moderate gains over encoder, extraction, few-shot, graph, and hypergraph baselines, with clearer improvements on tail and confusable labels. The method relies on event schemas or label descriptions to build label-side nodes, so tasks without such resources require schema induction or manual label descriptions before applying the model. Future work will study automatic schema construction and document-level argument aggregation for legal event extraction.

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